A probabilistic approach to logistics operations
What if delays were not random… but predictable?

Delays are not random.

Logistic Regression | Simulation | Interpretation

The real problem

- A delay is not just a late delivery — it’s a break in the operational flow.
- Delays impact costs, service level, planning, and people under pressure.
- There is no single cause. Delays emerge from multiple interacting factors.

No single cause. Multiple factors.

From events to probabilities

- Delays are usually treated as **isolated events:** it happened / it didn't.
- In reality, operations behave under ***uncertainty**, not certainty.
- The right question is not "Will it be delayed?" but "How likely is a delay?"

*Uncertainty is not noise. It's information

The data behind the problem

- The dataset represents typical logistics shipment conditions, inspired by real operational environments.
- It reflects the fusion of multiple operational data sources, not a single export.
- Data preparation required aligning information from:

1. ERP systems (orders, priorities, weights, distances)
2. TMS / routing tools (route type, number of stops)
3. Workforce data (operator experience, shifts)
4. WMS / dock systems (waiting and loading times)

- Building a usable table was a non-trivial effort and the true starting point of the model.

Modeling starts long before training a model.

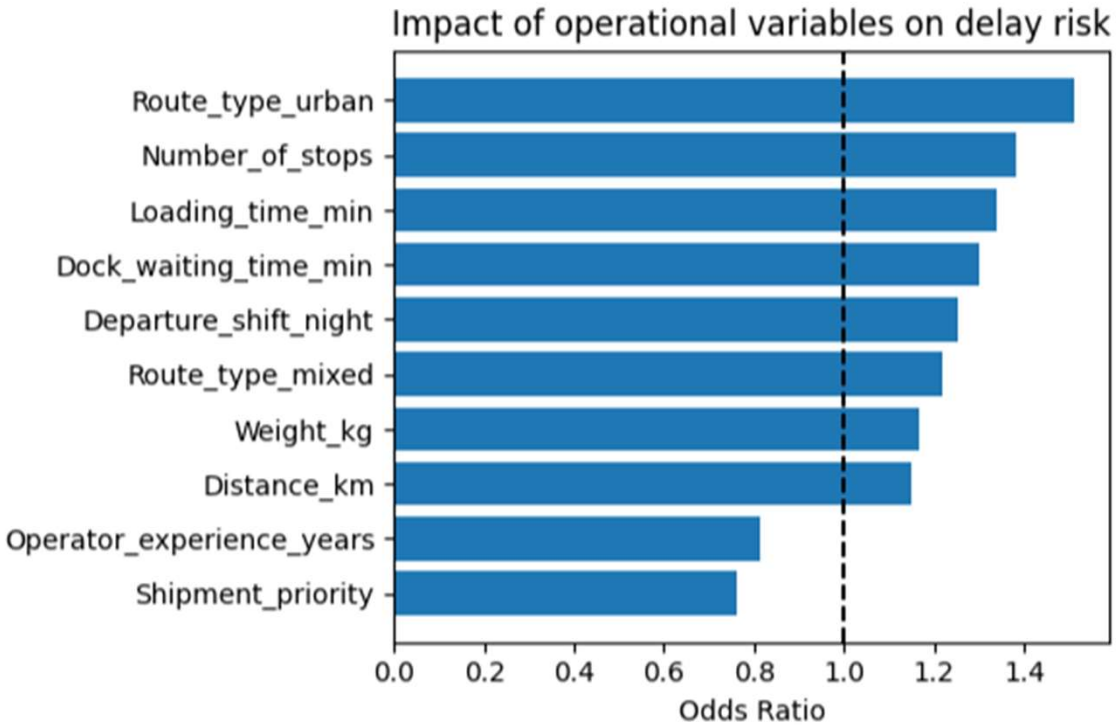
The model: Logistic Regression

- Logistic Regression was chosen for its simplicity, transparency, and interpretability.
- The model estimates the probability of delay, not just a yes/no outcome.
- Each variable contributes to risk in a measurable and explainable way.
- The focus is understanding why delays happen, not chasing perfect accuracy.

Operational inputs → Probability of delay → Decision support

Interpretability matters more than complexity.

What really drives delays



Accuracy score after 'model.predict': 0.706

Not surprises. Just quantified.

From insight to decision

- The simulator turns model coefficients into practical scenarios.
- Different operational conditions lead to different risk profiles.
- The model supports decision-making, not predictions carved in stone.

Same model. Different scenarios. Different risks.

01

```
=== Shipment Simulation ===
 Distance_km  Weight_kg  Number_of_stops
Route_type Departure_shift
         120        500                1
long           night

Model output:
Delay probability: 0.126 (12.6%)
Decision threshold: 0.50
Model classification: ON-TIME (0)
(0.1256720387499523, 0)
```

02

```
=== Shipment Simulation ===
 Distance_km  Weight_kg  Number_of_stops  Sh
Route_type Departure_shift
          50       2500                4
urban             day

Model output:
Delay probability: 0.960 (96.0%)
Decision threshold: 0.50
Model classification: DELAYED (1)
(0.9601463573149157, 1)
```

Probability is a guide, not a verdict.

Final reflection

This project is not about predicting the future with certainty.
It is about making **uncertainty visible.**
Logistic Regression does not provide answers but,
it provides better questions.

Not magic. Just probabilities.

Where f(x) meets Kaizen

LozanoLsa

Thank you!

Delays are not random events. They are the result of interacting processes.

Take it further

Feel free to download the full project:
the dataset, the notebook, and the simulator.

Change the data.
Break the code.
Ask different questions.

This model is just one snapshot in an ongoing process.
And that process never truly stops.

Because improvement has no finish line.